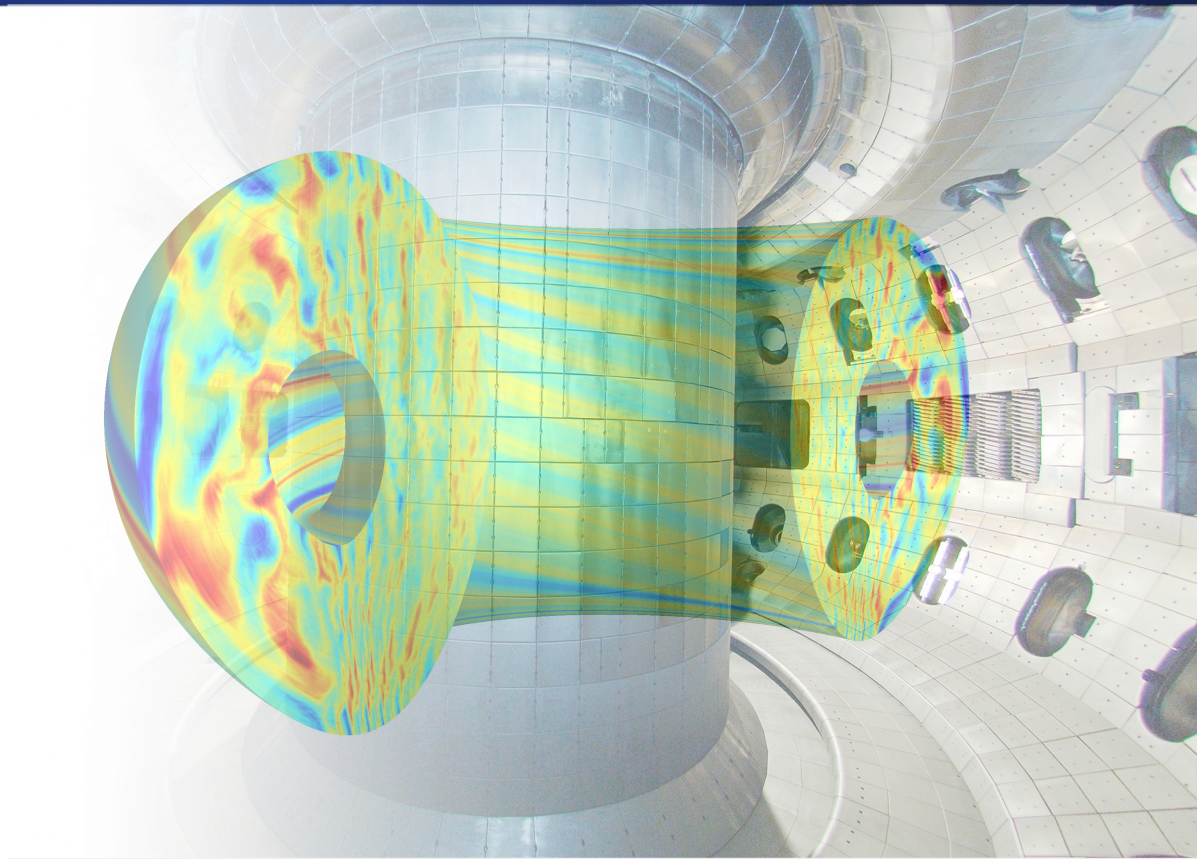
Plasma Turbulence and Transport: Workflows & Multi-Fidelity Methods

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Outline

- Plasma confinement in tokamaks
- Plasma transport
- Multi-fidelity methods for turbulence

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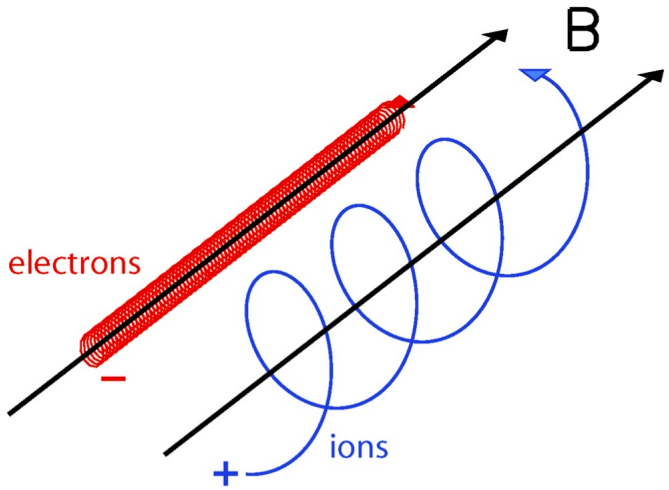
- **Plasma confinement in tokamaks**
- Plasma transport
- Multi-fidelity methods for turbulence

Confinement by Magnetic Fields

- Plasma can be confined by the gyration of charged particles in a magnetic field

- At $B = 10T$, $T = 10keV$
  - Larmor radius depends on B, m, Z, E
  - $\rho_e = 0.033\ mm$
  - $\rho_i = 1.5\ mm$

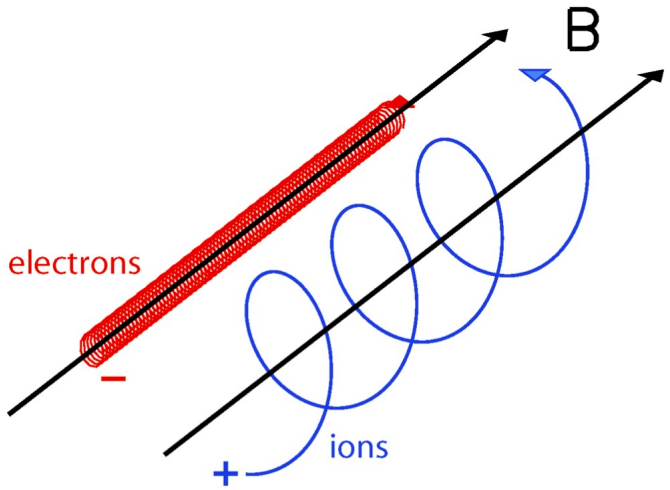
- Device size for $\tau_E \sim 1s$
  - $\tau_E \sim \frac{a^2}{\chi_i}$
  - $\chi_i \sim \frac{\Delta x^2}{\Delta t} \sim \frac{\rho_i^2}{(a/v_{ti})}$
  - $a \sim 1000\ \rho_i \sim 1.5m$



* Credit: C. Clauser

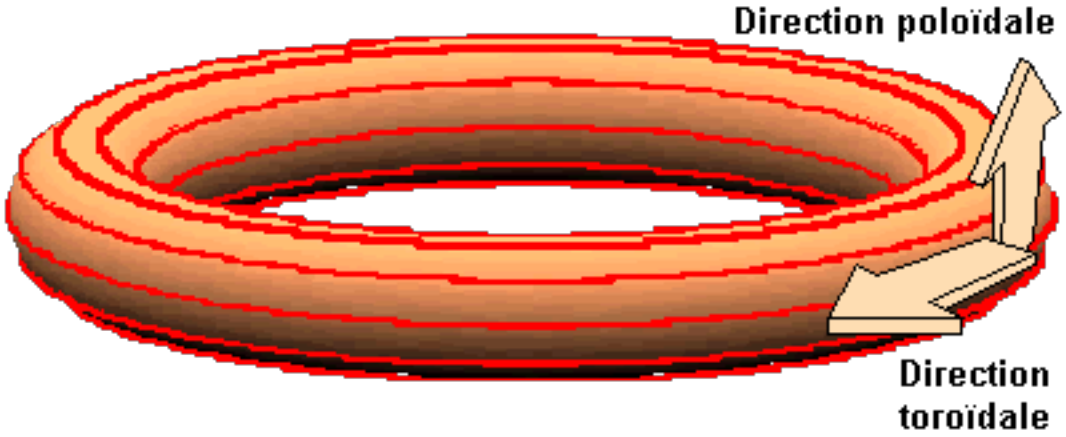
Confinement by Magnetic Fields

- Plasma can be confined by the gyration of charged particles in a magnetic field
- At the temperatures involved, ions are moving at ~1000km/s
- Avg time for ion to fuse is ~ 25 s
- So the end losses must be eliminated somehow



* Credit: C. Clauser

Toroidal configuration solves this problem!

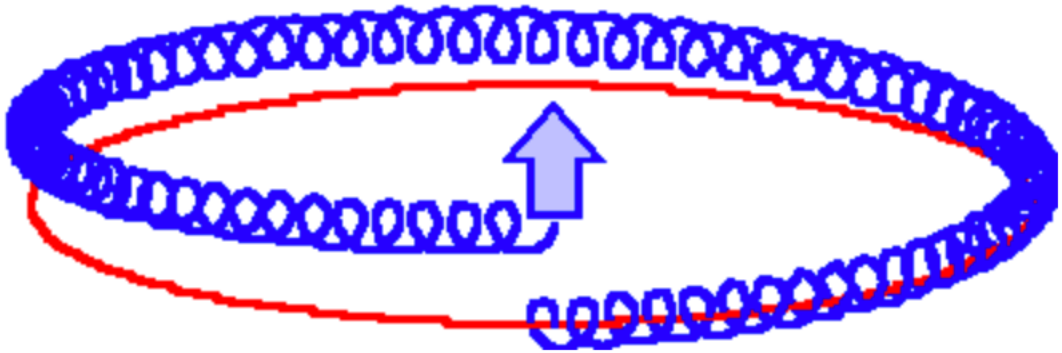


* Credit: CEA, IRFM

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Not quite …

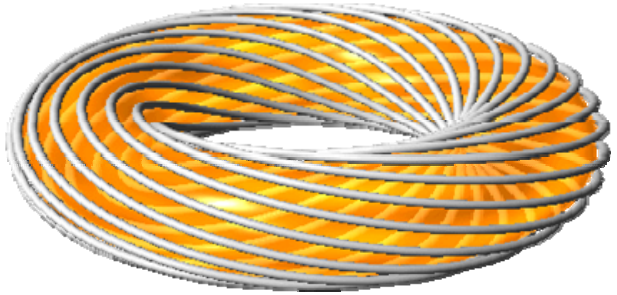
- Particles will undergo a slow transverse drift, linked to the magnetic field gradient and centrifugal force
- Direction of drift also depends on sign of charge

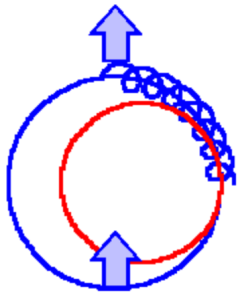


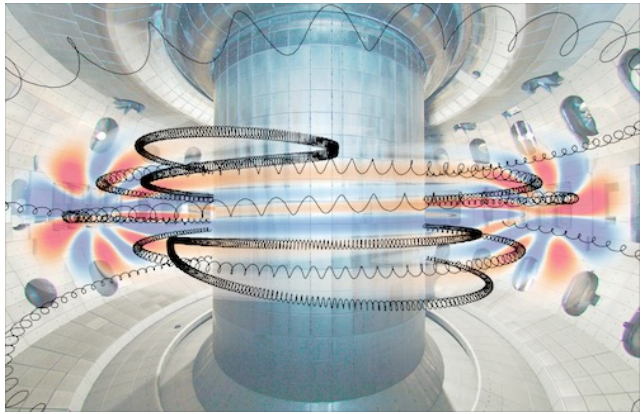
* Credit: CEA, IRFM

Tokamak

- Add a poloidal component to the toroidal magnetic field
  - Btor from **external toroidal field coils**
  - Bpol produced by a **toroidal current in the plasma**
  - Result: **"helical"** magnetic field lines
  - Vertical drift points in the same way but particle alternates btw top and bottom of flux surface
    - ½ time drift moves it away
    - ½ time drift brings it closer







Outline

- Plasma confinement in tokamaks
- **Plasma transport**
- Multi-fidelity methods for turbulence

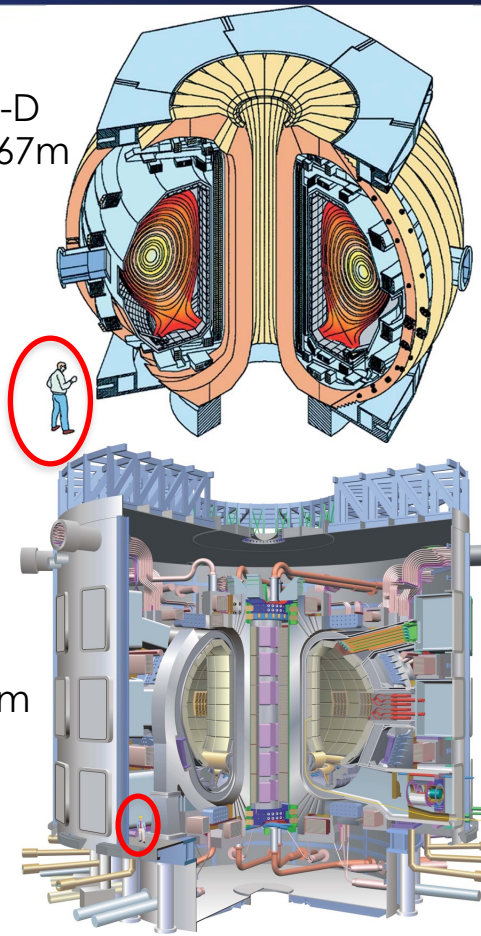
Energy confinement

Necessary plasma conditions for fusion:

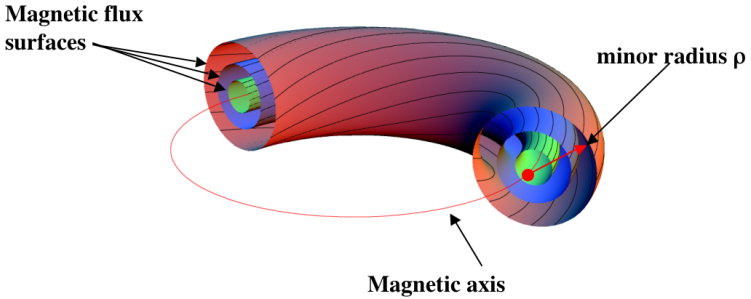
- **High density** (so ions collide frequently enough): $10^{20}/m^3$ (still 250,000 times thinner than air)
- **High temperature** (to overcome Coulomb repulsion): 10keV (150 million C)
- **Long energy confinement** (to maintain high temperature): ~3-4 s (how quickly heat leaks out through transport)

$$\tau_E \sim L_R^2/\chi$$

→ Must enlarge the device or reduce the heat transport

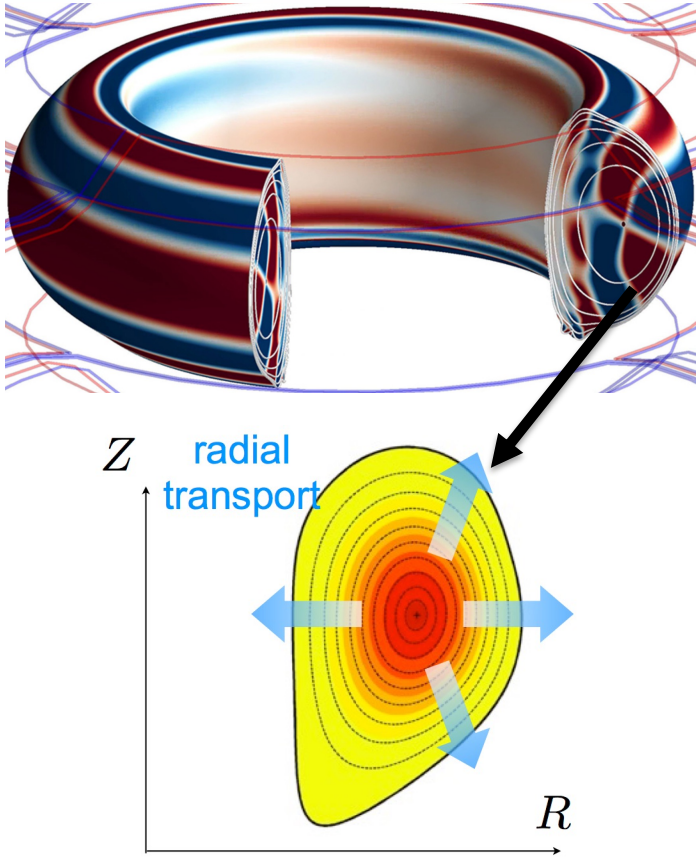


Plasma confinement is limited by slow particle and energy losses.



- Desired: **Hot, dense core**
- Soln: Magnetically confine plasma (tokamak)
- Field lines lie on closed nested surfaces (flux surfaces)

Plasma confinement is limited by slow particle and energy losses.



- **Radial transport << parallel transport**
- **Lower transport → better confinement**
- Follow radial evolution of (n,T) across flux surf
- **1D transport equations**

$$\frac{\partial \langle n_a \rangle}{\partial t} + \frac{1}{V'}\frac{\partial}{\partial r}(V'\Gamma_a) = S_{na}$$

$$\frac{\partial \langle n_a T_a \rangle}{\partial t} + \frac{1}{V'}\frac{\partial}{\partial r}(V'Q_a) = S_{Ta}$$

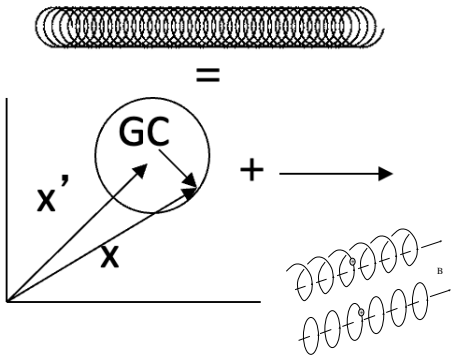
- **We use codes to compute the fluxes.**

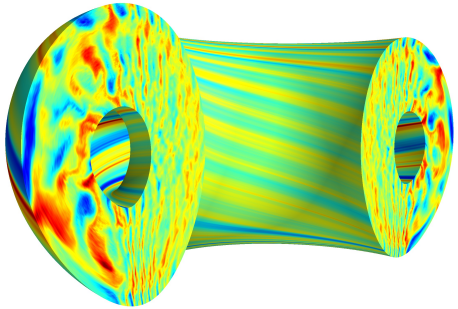
Plasma transport is generally described by two processes.

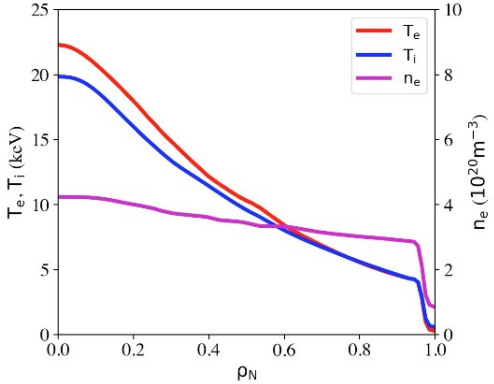
- **Neoclassical Transport**
  - Drift-Kinetic Simulations (**NEO**)
  - Due to Coulomb collisions of particles in toroidal magnetic field geometry

- **Turbulent Transport**
  - Gyrokinetic Simulations (**CGYRO**) or Models (**TGLF**)
  - Results from microturbulent fluctuations in the electric and magnetic fields, driven by gradients in temperature/density

A wide range of scales in space and time







Particle gyration

Ions spiral around B (mm, ns). Avg over gyromotion; rings of charge: 6D → 5D.

Turbulence (+NEO):

Small eddies (a few mm) that fluctuate on ~μs + Collisions in the curved fieldd

Profiles

Density & temperature across the whole device (1 m), evolving over seconds.

Each scale is ~1000 × apart ($\rho^* \approx 10^{-3}$) → Ccompute each piece separately and pass answers up the chain.

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Scale-Separated Simulations

Order	Question	Physics model	Code / source
0th	What is the background state?	Maxwellian with n(r), T(r) on flux surfaces + magnetic force balance	Profiles (measured or predicted) + EFIT geometry
1st	Given that state, how fast do particles and heat leak out?	Drift-kinetic (collisions) + gyrokinetic (turbulence)	NEO + CGYRO
2nd	How do the profiles evolve?	Transport equations	TGYRO

State → fluxes → updated state

2nd order closes the loop: it updates the 0th-order profiles.

Why it works

Very different time scales ($\rho^* \ll 1$), so each level can be solved separately.

The transport equation is a conservation law.

$$\frac{\partial \langle \boldsymbol{n_a T_a} \rangle}{\partial t} + \frac{1}{V'} \frac{\partial}{\partial r} (V' \boldsymbol{Q_a}) = \boldsymbol{S_{Ta}}$$

rate of change of stored energy

net heat flowing out through the surface

heating

Steady state

Heat leaving each flux surface = all the power deposited inside it.

That fixes a target flux $Q_{target}(r)$ at every radius.

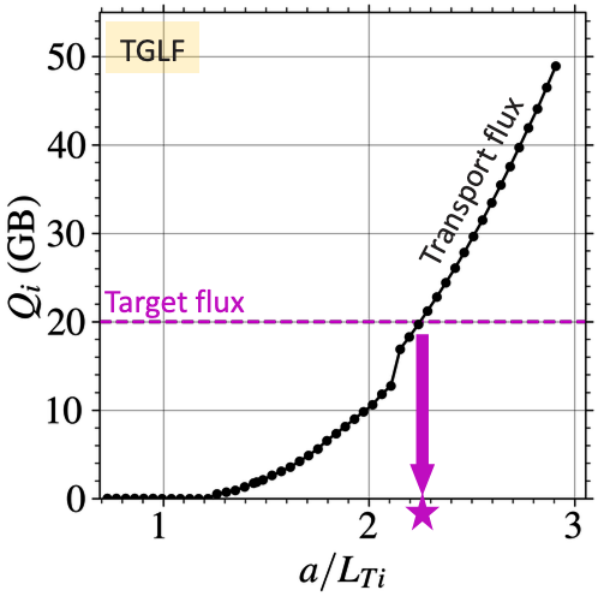
The unknown: the heat flux Q

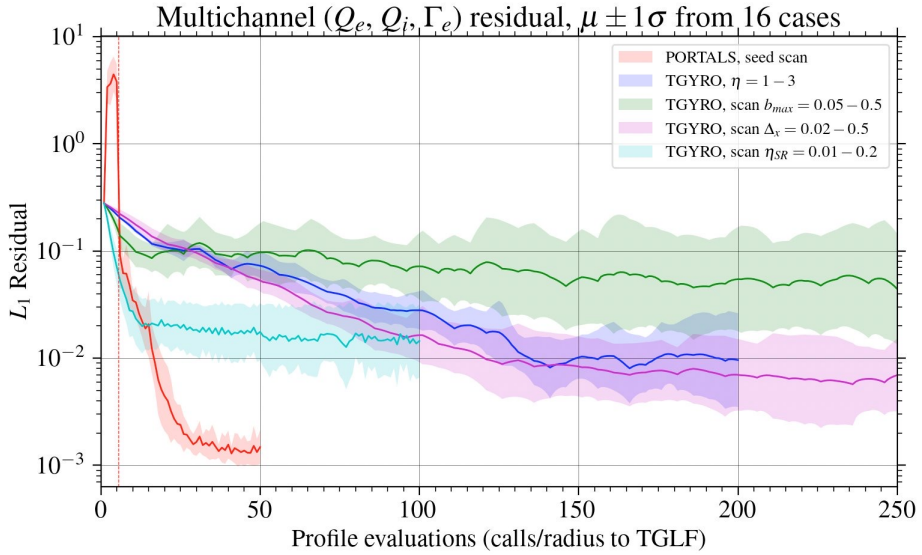
$$Q_a = Q_a^{neo} + Q_a^{turb}$$

Depends nonlinearly on the local gradients. Computed by NEO (collisions) + CGYRO (turbulence).

Predicting Profiles: Nonlinear Root Finding with an Expensive function

- Solve $\mathbf{Q_{model}(gradients) = Q_{target}}$ at every r, then integrate inward to get T(r)
- **Stiff:** the flux rises steeply past a threshold, so the root sits on a steep curve
- Each evaluation of f = one CGYRO run (HPC); a solve needs hundreds





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The flux calculation is a function: Q = f(x)

Inputs x

20+ dimensionless numbers

- Gradients: a/L_T, a/L_n
- Temperature ratio: T_i/T_e
- Density ratio: n_i/n_e
- Magnetic geometry: q, shear, shape (κ, δ, …)
- Collisionality: ν
- Plasma pressure: β
- Impurities



CGYRO

nonlinear gyrokinetic simulation

Many hours on GPUs



Output y

Q_i, Q_e, Γ

time-averaged ion & electron heat flux, particle flux

- **Q = f(x): expensive, nonlinear, and noisy**
- Need to evaluate y **faster yet accurately**

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- Plasma confinement in tokamaks
- Plasma transport
- **Multi-fidelity methods for turbulence**

Multi-Fidelity Hierarchy for Gyrokinetic Turbulence

High-Fidelity: Nonlinear Gyrokinetics

- First-principles simulation of turbulent transport (**CGYRO**)
- Requires HPC/exascale resources → expensive!
- Hours to days per simulation
- Ground truth for physics validation

Medium-Fidelity: Quasi-Linear Models

- Exploit weak turbulence, knowledge of linear physics (**TGLF**)
- Seconds on laptop
- Used for routine analysis and in integrated modeling

Low-Fidelity: Neural Network Surrogates

- Train on nonlinear + quasi-linear database
- Milli-second evaluation time
- Time-dep integrated modeling
- Uncertainty quantification included

Physics-based Quasi-Linear Transport Models

- The Challenge: Fast enough computational time for turbulent fluxes to enable steady-state transport modeling
- Accurate enough to extrapolate to burning plasmas
- Combine: Linear GK + "Saturation Rule" calibrated w/ nonlinear GK

$$Q \sim W^L \; I$$

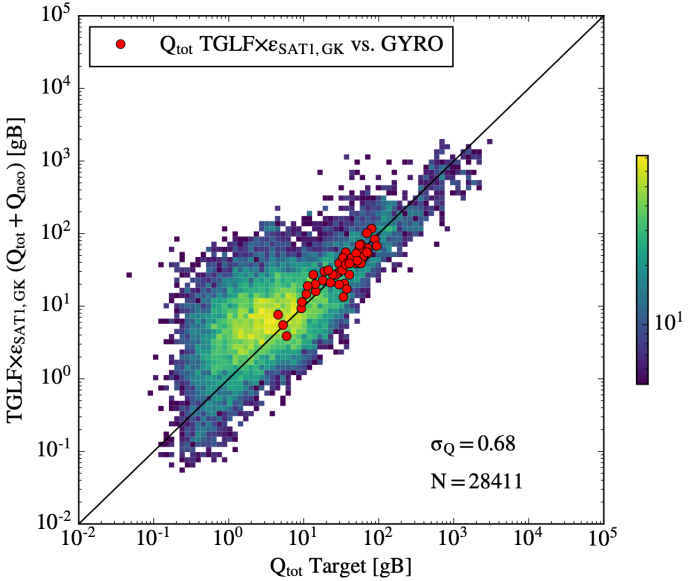
Linear weight
→ Computed in real time from linear solver (relatively fast)

Intensity spectrum
→ "prescribed" by saturation rule

Saturation Rules: Calibrated to nonlinear GK

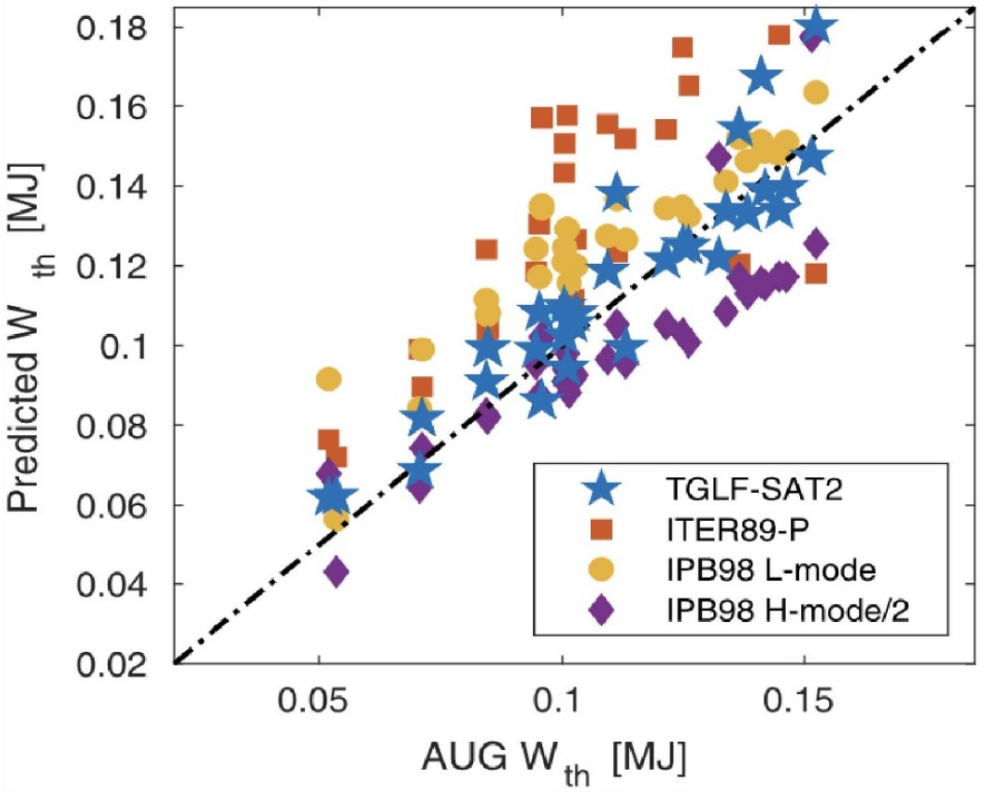
- ***TGLF SAT0-SAT3***
- Physics-based formula
- Low number of **free coefficients**
- Coeff determined by **calibration to nonlinear GK** by nonlinear least-squares fitting to a database of GK fluxes from CGYRO

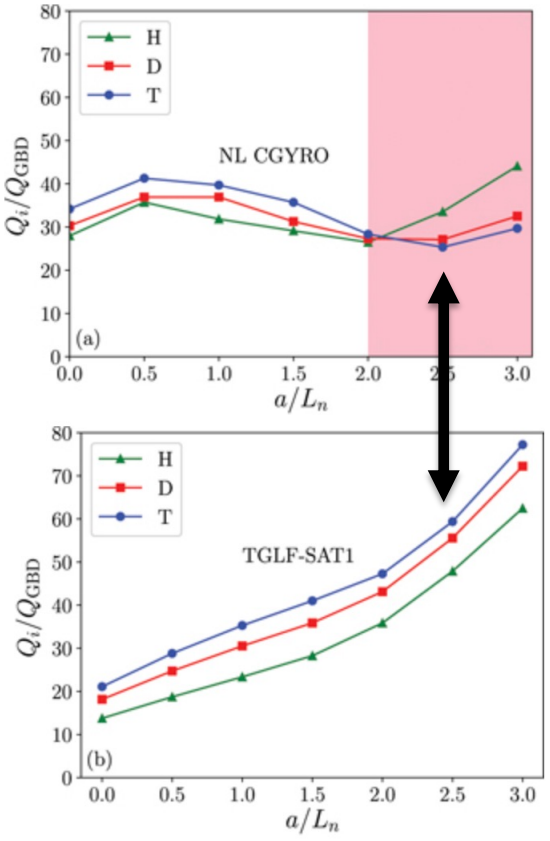
$$I_{k_y}^{\text{model}} = C_{\text{n}} \left(\frac{\gamma_{\text{model}} \left(\frac{\gamma_{12}}{\gamma_1} \right)^{\text{exp1}}}{k_y^2 \left[1 + \left(a_y \frac{k_{x0_e}}{k_y} \right)^2 \right]} \max \left[\sqrt{\frac{k_y}{k_{\text{ETG}}}}, 1.0 \right] \right)^2$$



Review article: G. Staebler, C. Bourdelle, J. Citrin, R. Waltz, "Quasilinear theory and modelling of gyrokinetic turbulent transport in tokamaks", NF 64, 103001 (2024).

QL models exceed empirical scaling laws in prediction but there are "gaps" where QL is inaccurate

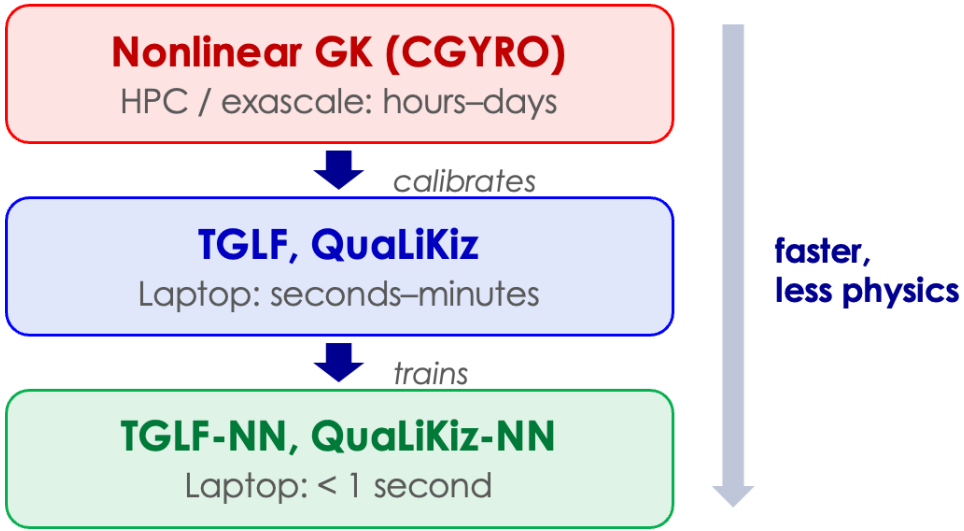


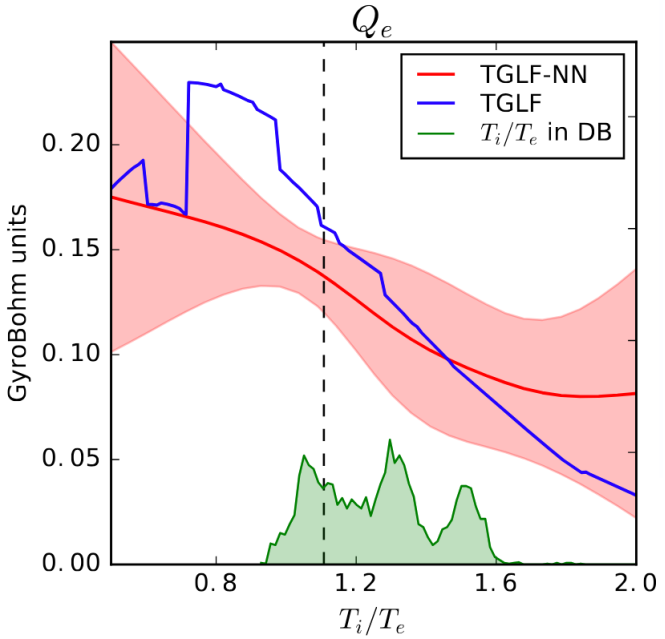


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NN surrogate models for QL codes

- TGLF-NN (Meneghini, 2017)
- QuaLiKiz-NN (van de Plassche, 2020)





Regularization smooths out discontinuities which affects convergence of transport solvers

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Projects: Learning turbulent fluxes from data

1A Neural-net surrogate for TGLF

Goal: learn TGLF (~15 inputs → fluxes) in milliseconds

Data: TGLF database, inputs derived from exp

- Find correlations among input parameters
- Compare regression, trees, and neural nets

1B Learning the saturation rule

Goal: improve TGLF's saturation rule with CGYRO data

Data: CGYRO fluxes + TGLF linear outputs

- Reproduce the "SAT" fit (simple regression)
- Try NN approaches for fitting the coefficients

Shared challenges

- Correlated inputs: real plasmas fill only a thin slice of input space
- Sparse GK data
- "Bad"/anomalous data
- Uncertainty quantification: an error bar on every prediction
- Outside the data or large uncertainty